

RESEARCH ARTICLE

Fault Tolerant Control of a Quadrotor Unmanned Aerial Vehicle Based on Active Disturbance Rejection Control and Two-Stage Kalman Filter

YUSHENG DU, PENG HUANG, YUN CHENG^{ID}, YUNLEI FAN, AND YINLONG YUAN

School of Electrical Engineering, Nantong University, Nantong 226019, China

Corresponding author: Yun Cheng (chengyun@ntu.edu.cn)

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ABSTRACT This paper addresses the fault tolerant control problem of a quadrotor unmanned aerial vehicle (UAV) with the actuator loss-of-effectiveness fault and measurement noise. To this end, a cascade active disturbance rejection control (ADRC) strategy with a two-stage Kalman filter (TSKF) is proposed. The TSKF can estimate the loss-of-effectiveness fault factor and the state of the UAV in a noisy environment. Considering that unknown disturbance will influence the estimation accuracy of the TSKF, an extended state observer (ESO) is designed in the ADRC controller to further compensate for the influence of the factor estimation error. In addition, the convergence of the ESO and the stability of the closed-loop system are analyzed. Finally, the proposed strategy is compared with several traditional control methods in the simulation experiments. The simulation results show that the proposed strategy has good stability and disturbance rejection performance, and it has a clear potential in the application of the UAV flight control.

INDEX TERMS Quadrotor unmanned aerial vehicle, active disturbance rejection control, two-stage Kalman filter, loss-of-effectiveness fault, measurement noise.

I. INTRODUCTION

A quadrotor UAV is a type of multi-rotor aircraft that consists of four rotors driven by electric motors. Compared to other flying vehicles, quadrotor UAVs have the characteristics of small volume, light weight and flexible control. They can perform tasks in narrow or dangerous environments. Therefore, quadrotor UAVs are widely used in aerial photography, mapping, search and rescue, agriculture, scientific research and other fields [1], [2]. As the quadrotor UAV is a typical underactuated mechanical system, its nonlinear, multi-input, multi-output, strong coupling, input saturation and other characteristics make its stable flight control challenging [3], [4], [5], [6]. Additionally, the aircraft's environmental awareness and autonomous decision-making abilities also demand higher standards for its control. Researchers have developed advanced motion control algorithms to overcome these challenges, and continuously improve the performance

and stability of quadrotor UAVs to meet the application needs of different fields. The increasing use of quadrotor UAVs in various fields has led to actuator failure becoming a problem that cannot be ignored. The actuator includes motors, electrical regulators and rotors. Once the actuator fails, it will lead to a significant decline in the flight performance of the UAV and even cause serious accidents such as crashes [7]. How to ensure the stability and safety of quadrotor UAVs have become an urgent issue to be solved.

By developing a flight fault tolerant control system, the fault tolerance and self-healing ability of quadrotor UAVs for actuator failures are effectively improved, and the impact of actuator failures on flight safety is reduced. Fault tolerant control systems are usually composed of two parts: fault diagnosis and control reconfiguration [8]. Fault diagnosis is responsible for detecting and recognizing actuator failures, including motor faults and rotor faults. Once a fault is detected, control reconfiguration will be initiated to re-plan the flight trajectory and control commands of the UAV,

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in order to adapt to the fault condition and ensure the performance and stability of the system.

Fault diagnosis can be divided into two types: model-based and data-based [9]. Model-based fault detection methods are widely used in current quadrotor UAVs fault diagnosis, which mainly rely on parameter estimation, parity equation and state observers. In [10] and [11], a fault detection scheme based on observers is designed to estimate the actuator failures under disturbances. In [12], considering the sudden motor loss-of-effectiveness fault of UAV, adaptive technology is used to estimate the fault parameters online to restrain the effects of motor loss-of-effectiveness fault in the tracking loop. The above scheme can achieve good fault detection performance under the presence of external disturbances. Considering that there are more than one factor affecting the performance of UAVs in a complex environment, such as system nonlinearity and parameter uncertainty, the effectiveness of the above scheme will be greatly reduced. At this point, it is necessary to improve the fault diagnosis scheme. In [13], a fault diagnosis scheme composed of a comprehensive observer and a set of fault isolation estimators based on UAV actuators is proposed to deal with system nonlinearity and external disturbances. In [14], a robust H_∞ observer is proposed to enable fault and state estimation of a quadrotor UAV with actuator failures, disturbances and parameter uncertainties. Reference [15] proposes a polynomial controller with compensations for actuator faults, where a nonlinear high-gain observer-based estimator is presented to provide the estimations of the unmeasurable signals and faults.

A data-driven fault diagnosis scheme does not rely on a system model. It works with a large amount of historical data. An indirect neural network based adaptive control scheme is developed in [16], for handling modeling uncertainties and actuator faults. But the weight update of the neural network is slow and the computation amount is large, resulting in the easy influence of sudden faults for UAVs. Therefore, a novel neural network fault detection technique is developed, which uses an extended Kalman filter to update the weighted parameters of the neural network [17]. In comparison to traditional fault detection using neural networks, it exhibits superior performance.

With fault diagnosis, fault tolerant control can effectively compensate for the impacts of faults to ensure system stability and security. Fault tolerant control of quadrotor UAVs is mainly divided into two methods: active fault tolerance and passive fault tolerance. In active fault tolerant control, references [18], [19], [20] use sliding mode control to achieve fault tolerant control of a quadrotor UAV in the event of the actuator failure. In [21], a fault tolerant control approach based on gain scheduling is designed in the framework of structured H_∞ synthesis for a quadrotor UAV with actuator failures. The effectiveness and robustness of the system have been verified through experiments with actuator failure conditions. Compared to active fault tolerant control, passive fault tolerant control does not require diagnosing faults and does not require reconfiguring the controller. There are also numerous fault tolerant control methods, such as fuzzy neural

network control [22], sliding mode control [23] and adaptive control [24]. The above methods can provide fault control in a timely and effective way when the actuator fails. However, if there are external disturbances and measurement noise at the same time, it will influence the control performance, and the problem will become more complicated.

Kalman filtering techniques are an effective solution for the measurement noise problem. When there is random bias in the system, two-stage Kalman filtering can be used to estimate the optimal solution of the dynamic set [25], [26], [27]. For quadrotor UAVs, the actuator failure can be seen as a system random bias, and two-stage Kalman filtering can be used to estimate the actuator failure. However, two-stage Kalman filtering techniques require accurate models, and when the system has unknown external disturbance or uncertainty, the estimation will have bias, thus increasing the difficulty of the fault tolerant controller design.

Active disturbance rejection control (ADRC), which inherits many advantages of PID controllers, is an effective way to deal with model uncertainty and disturbances [28]. It is regarded as an emerging robust control technology [29], [30] with a wide application in UAV flight control. In [31], ADRC is combined with a disturbance observer to estimate real-time wind disturbances, effective load disturbances and propeller fault disturbances. In [32], a fault tolerant flight control method for a quadrotor UAV under actuator failure and external wind disturbances is proposed. The control method is based on ADRC and spatial radial basis function neural networks, which has been experimentally demonstrated to be effective. It can be seen that ADRC has a unique application prospect in UAV fault tolerant control.

Inspired by the above-mentioned discussion, this paper combines ADRC with a two-stage Kalman filter to realize the fault tolerant control of a quadrotor UAV with the actuator loss-of-effectiveness fault and measurement noise. A two-stage Kalman filter is designed to estimate the loss-of-effectiveness fault factor, which offers the filtering data and controller parameters for ADRC. ESO is combined with the TSKF. ESO ensures that the TSKF works when there are biases in the system or uncertainties in the model, and the input to ESO is filtered by the TSKF to attenuate the effect of noise on ESO. Additionally, ADRC with extended state observer (ESO) is utilized to ensure effective control even when there is a bias in the Kalman filter estimation. The main contributions of this paper are summarized as follows.

(1) The dynamic model of a quadrotor UAV with the actuator loss-of-effectiveness fault is developed and a two-stage Kalman filter is designed to estimate the loss-of-effectiveness fault factor in the presence of measurement noise.

(2) A cascaded ADRC structure based on a two-stage Kalman filter is designed to address the fault-tolerant flight control problem of quadrotor UAVs. The two-stage Kalman filter provides filtered outputs and estimated loss-of-effectiveness fault factors. The ADRC controller timely compensates for the control input based on the estimated loss-of-effectiveness fault factor, thereby ensuring a stable flight.

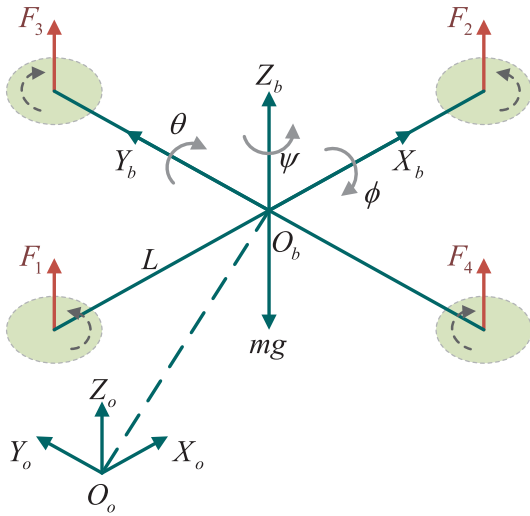


FIGURE 1. The coordinate system and the structure of a quadrotor UAV.

(3) The convergence of ESO and the stability of the closed-loop system are analyzed, and the proposed method is compared with several traditional methods through simulation experiments to verify its effectiveness.

The structure of this paper is arranged as follows. The control problem is discussed in Section II. Section III presents the design procedure of the two-stage Kalman filter. In Section IV, an ADRC control scheme based on two-stage Kalman filtering is proposed and the system stability is analyzed. In Section V, the proposed scheme is simulated and compared with several traditional methods. The conclusion is presented in the final section.

II. PROBLEM FORMULATION

A. DYNAMIC MODEL OF THE QUADROTOR UAV

A quadrotor UAV is an underactuated system with six degrees of freedom. It consists of four rotors that are symmetrically distributed, and the center of gravity is located at the geometric center. Each motor generates a thrust of T_i . There is a first-order linear relationship between the corresponding PWM input and its transfer function is

$$T_i = K \frac{\omega}{s + \omega} u_i, i = 1, 2, 3, 4 \quad (1)$$

where K is a positive gain and ω is the motor bandwidth. When $u_i = 0$, the corresponding motor provides zero thrust.

In order to obtain the dynamic model of a quadrotor UAV, it is necessary to first establish the body coordinate system $\{B\}$ and the ground coordinate system $\{O\}$, with its coordinate system and the structure shown in Figure 1. O_o is chosen as the origin of the ground coordinate system, X_o , Y_o and Z_o are three directions of the ground coordinate system. O_b is the center of the body coordinate system, X_b , Y_b and Z_b are three axis of the body coordinate system respectively. The x axis points to the right front direction, the y axis points to the left front.

In Figure 1, it has six degrees of freedom: three position variables (x, y, z) and three attitude variables (θ, ϕ, ψ) , θ

represents the pitch angle, ϕ represents the roll angle and ψ represents the yaw angle. The torque τ_i generated by the motor when it rotates is opposite to the direction of the motor torque. The thrust T_i generated by the i -th propeller always points upwards in the z axis direction, parallel to the axis of rotation of the motor. Thrust T_i and torque τ_i generate the lift in the z axis direction and torques about the x , y and z axis. The relationship between the lift/torque and the thrust is shown as follows:

$$\begin{cases} u_z = T_1 + T_2 + T_3 + T_4 \\ u_\theta = L(T_1 - T_2) \\ u_\phi = L(T_3 - T_4) \\ u_\psi = \tau_1 + \tau_2 - \tau_3 - \tau_4 \end{cases} \quad (2)$$

where L is the distance from the rotor to the center of the quadrotor UAV body, u_z , u_θ , u_ϕ and u_ψ represent the total lift, pitch torque, roll torque and yaw torque of a quadrotor UAV respectively. The torque of the electric motor is directly related to the thrust it produces. The relationship is $\tau_i = K_\psi T_i$, where K_ψ represents the driving torque factor. According to equation (1), we can take $T_i \approx K u_i$, and equation (2) can be expressed as

$$\begin{bmatrix} u_z \\ u_\theta \\ u_\phi \\ u_\psi \end{bmatrix} = \begin{bmatrix} K & K & K & K \\ KL & -KL & 0 & 0 \\ 0 & 0 & KL & -KL \\ KK_\psi & KK_\psi & -KK_\psi & -KK_\psi \end{bmatrix} u \quad (3)$$

where $u = [u_1 \ u_2 \ u_3 \ u_4]^T$.

According to the Euler-Lagrange formulation, a dynamic model for the quadrotor UAV [33] is established as follows:

$$\begin{cases} m\ddot{x} = u_z(\cos \phi \sin \theta \cos \psi + \sin \phi \sin \psi) \\ m\ddot{y} = u_z(\cos \phi \sin \theta \sin \psi - \sin \phi \cos \psi) \\ m\ddot{z} = u_z(\cos \phi \cos \theta) - mg \\ J_1\ddot{\theta} = u_\theta + d_1 \\ J_2\ddot{\phi} = u_\phi + d_2 \\ J_3\ddot{\psi} = u_\psi + d_3 \end{cases} \quad (4)$$

where x , y and z are the coordinates of the vehicle's centroid in the earth frame, d_1 , d_2 and d_3 are external disturbances in the attitude loop. The descriptions and values of parameters in equation (4) are given in Table 1. If the quadrotor UAV is in a state without yaw deviation ($\psi=0$) and the roll angle ϕ and pitch angle θ are relatively small or hovering, then the roll angle and pitch angle can be defined as zero, and a simplified motion linear model can be obtained as

$$\begin{cases} \ddot{x} = \theta g \\ \ddot{y} = -\phi g \\ \ddot{z} = \frac{u_z}{m} - g \\ J_1\ddot{\theta} = u_\theta + d_1 \\ J_2\ddot{\phi} = u_\phi + d_2 \\ J_3\ddot{\psi} = u_\psi + d_3 \end{cases} \quad (5)$$

TABLE 1. Parameters of the quadrotor UAV flight system.

Parameter	Description	Value	Unit
L	Length between the center of the UAV and the rotor	0.2	m
K	Thrust gain	175	
K_ψ	Thrust-to-moment gain	0.023	
ω	Motor bandwidth	15	rad/s
m	Mass	1.42	kg
g	Gravity	9.81	m/s ²
J_1	Rotational inertia of the x axis	0.03	kg · m ²
J_2	Rotational inertia of the y axis	0.03	kg · m ²
J_3	Rotational inertia of the z axis	0.04	kg · m ²

B. ACTUATOR FAILURE DESCRIPTION

A discrete linear time-varying state space model is used to describe the quadrotor UAV system according to equation (5), and the representation is shown as

$$\begin{cases} x_{k+1} = A_k x_k + B_k u_k + w_k^x \\ y_{k+1} = C_k x_{k+1} + v_{k+1} \end{cases} \quad (6)$$

where the subscript k represents the value at the moment k . x_k , u_k and y_k are the state variables, control input variables and output variables respectively, A_k , B_k and C_k stand for the state matrix, input matrix, and output matrix respectively. w_k^x is the system state noise, v_k is the measurement noise of the sensor, w_k^x and v_k are mutually independent.

When the quadrotor UAV actuator fails, in order to facilitate the detection and estimation of the loss-of-effectiveness fault, the loss-of-effectiveness fault factor is described as follows:

$$b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}; 0 \leq b_i \leq 1; i = 1, 2, 3, 4 \quad (7)$$

where b is the vector composed of the actuator loss-of-effectiveness fault factor, when $b_i = 0$ or $b_i = 1$, it means that the i -th actuator is either completely fine or completely damaged. The dynamic model of the quadrotor UAV with actuator failure can be rewritten as the following form:

$$\begin{cases} x_{k+1} = A_k x_k + B_k u_k - B_k U_k b_k + w_k^x \\ y_{k+1} = C_k x_{k+1} + v_{k+1} \end{cases} \quad (8)$$

where $U_k = \text{diag}[u_{1k} \ u_{2k} \ u_{3k} \ u_{4k}]$.

The focus of this paper is to design a controller for a quadrotor UAV by utilizing the state space model with actuator failure, which is described in equation (8). The primary goal of the control system is to maintain stable flight even in the event of an actuator loss-of-effectiveness fault in the quadrotor UAV.

III. TWO-STAGE KALMAN FILTER

The Kalman filter is widely used as an optimal filtering method, based on a linear least mean square criterion. The TSKF, developed from the Kalman filter, has numerous applications in parameter bias estimation and state estimation. In this paper, the TSKF can also be utilized in fault diagnosis.

Compared with the conventional Kalman filter, the TSKF decomposes the system state filter into two low-dimensional filters, namely the “bias-free” filter and the “bias” filter. The results of the “bias-free” filter are corrected by the results of the “bias” filter to obtain the final state estimation. In this paper, TSKF is used to estimate the loss-of-effectiveness fault factor and the state of the UAV. The actuator failures are treated as enhanced random bias in the TSKF structure.

Based on equation (8), the bias augmented discrete linear time-varying state-space model for a quadrotor UAV is given as

$$x_{k+1} = A_k x_k + B_k u_k - B_k U_k b_k + w_k^x \quad (9)$$

$$b_{k+1} = b_k + w_k^b \quad (10)$$

$$y_{k+1} = C_k x_{k+1} + v_{k+1} \quad (11)$$

The system state estimate $\tilde{x}_{k|k}$ and the corresponding error covariance $\tilde{P}_{k|k}^x$ at k moment are known, and the bias estimation $\hat{b}_{k|k}$ and the corresponding error covariance $P_{k|k}^b$ at k moment are also known. The construction of the TSKF is divided into a “bias-free” filter and a “bias” filter [25].

First, the bias is ignored and the “bias-free” filter is constructed in the following step 1 to step 4.

Step 1: The system state estimation $\tilde{x}_{k|k}$ is predicted according to equation (9), and the one-step prediction estimation $\tilde{x}_{k+1|k}$ and their error covariances $\tilde{P}_{k+1|k}^x$ are obtained as follows:

$$P_{k+1|k}^b = P_{k|k}^b + Q_k^b \quad (12)$$

$$\tilde{x}_{k+1|k} = A_k \tilde{x}_{k|k} + B_k u_k + W_k \hat{b}_{k|k} - V_{k+1|k} \hat{b}_{k|k} \quad (13)$$

$$\tilde{P}_{k+1|k}^x = A_k \tilde{P}_{k|k}^x A_k^T + Q_k^x + W_k P_{k|k}^b W_k^T - V_{k+1|k} P_{k+1|k}^b V_{k+1|k}^T \quad (14)$$

where Q_k^x is the covariance matrix of the system state noise w_k^x , Q_k^b is the covariance matrix of the bias Gaussian white noise w_k^b , W_k can be obtained from the coupled equation $W_k = A_k V_{k|k} - B_k U_k$, $V_{k+1|k}$ is the compensation factor of the bias estimation of the “bias” filter to the estimate of the system state in the “bias-free” filter, it can be obtained from $V_{k+1|k} = W_k P_{k|k}^b (P_{k+1|k}^b)^{-1}$.

Step 2: The prediction of $\hat{y}_{k+1|k}$ and the filter gain matrix $\tilde{K}_{k+1}^x$ for bias-free filtered state estimation are calculated.

$$\hat{y}_{k+1|k} = C_{k+1} \tilde{x}_{k+1|k} \quad (15)$$

$$\tilde{K}_{k+1}^x = \tilde{P}_{k+1|k}^x C_{k+1}^T (C_{k+1} \tilde{P}_{k+1|k}^x C_{k+1}^T + R_{k+1})^{-1} \quad (16)$$

where R_{k+1} is the covariance matrix of the measurement noise v_k .

Step 3: The one-step prediction estimation $\tilde{x}_{k+1|k}$ in step 1 is corrected according to the $\hat{y}_{k+1|k}$ obtained from step 2 with the bias-free filter gain matrix $\tilde{K}_{k+1}^x$.

$$\tilde{x}_{k+1|k+1} = \tilde{x}_{k+1|k} + \tilde{K}_{k+1}^x (y_{k+1} - \hat{y}_{k+1|k}) \quad (17)$$

Step 4: The error covariance of the one-step prediction estimation $\tilde{P}_{k+1|k}^x$ obtained in step 1 is updated according to the bias-free filter gain matrix $\tilde{K}_{k+1}^x$ obtained from step 2.

$$\tilde{P}_{k+1|k+1}^x = (I - \tilde{K}_{k+1}^x C_{k+1}) \tilde{P}_{k+1|k}^x \quad (18)$$

Next, an optimal bias filter is constructed, including a bias filter as well as a compensation filter. The compensation filter is used to compensate for the bias-free filter, while the bias filter is constructed in the following step 5 to step 8.

Step 5: The system bias estimation $\hat{b}_{k|k}$ is first predicted to obtain a one-step prediction bias estimation $\hat{b}_{k+1|k}$ as follows:

$$\hat{b}_{k+1|k} = \hat{b}_{k|k} \quad (19)$$

Step 6: The bias filter gain matrix K_{k+1}^b can be obtained based on the residual $\tilde{r}_{k+1}$ and its covariance $\tilde{S}_{k+1}$ of the sensor observation information and the one-step prediction estimation $\tilde{x}_{k+1|k}$ at k moment obtained in step 1.

$$\tilde{r}_{k+1} = y_{k+1} - C_{k+1}\tilde{x}_{k+1|k} \quad (20)$$

$$\tilde{S}_{k+1} = C_{k+1}\tilde{P}_{k+1|k}^x C_{k+1}^T + R_{k+1} \quad (21)$$

$$K_{k+1}^b = P_{k+1|k}^b H_{k+1}^T (H_{k+1|k} P_{k+1|k}^b H_{k+1}^T + C_{k+1}\tilde{P}_{k+1|k}^x C_{k+1}^T + R_{k+1})^{-1} \quad (22)$$

where $H_{k+1|k}$ can be obtained from the coupled equation $H_{k+1|k} = C_{k+1}V_{k+1|k}$.

Step 7: The one-step prediction bias estimation $\hat{b}_{k+1|k}$ in step 5 can be corrected according to the bias filter gain matrix K_{k+1}^b obtained in step 6. The bias estimation of the bias filter at k moment is obtained as

$$\hat{b}_{k+1|k+1} = \hat{b}_{k+1|k} + K_{k+1}^b (y_{k+1} - C_{k+1}\tilde{x}_{k+1|k} - H_{k+1|k}\hat{b}_{k|k}) \quad (23)$$

Step 8: The $P_{k+1|k}^b$ obtained in step 1 is updated according to the bias filter gain matrix K_{k+1}^b obtained in step 6.

$$P_{k+1|k+1}^b = (I - K_{k+1}^b H_{k+1|k}) P_{k+1|k}^b \quad (24)$$

Finally, the compensation filter is constructed. The optimal bias filter can be obtained by the linear combination of bias filter and the compensation filter. The final system state estimation is obtained from the “bias-free” filter state estimation corrected by the “bias” filter estimation in the following step 9 to step 10.

Step 9: The compensation factor $V_{k+1|k+1}$ is calculated to the estimated system state in the “bias-free” filter.

$$V_{k+1|k+1} = V_{k+1|k} - \tilde{K}_{k+1}^x H_{k+1|k} \quad (25)$$

where $P_{k+1|k}^b$ is the result of the calculation in step 1, $\tilde{K}_{k+1}^x$ is the filter gain matrix of the “bias-free” filter in step 2.

Step 10: Based on the compensation factor in step 9, the bias estimation $\hat{b}_{k+1|k+1}$ from the “bias” filter process is used to compensate for the system state estimation $\tilde{x}_{k+1|k+1}$ from the “bias-free” filter. The final state estimation and its error covariance matrix are obtained as follows:

$$\hat{x}_{k+1|k+1} = \tilde{x}_{k+1|k+1} + V_{k+1|k+1}\hat{b}_{k+1|k+1} \quad (26)$$

$$P_{k+1|k+1} = \tilde{P}_{k+1|k+1} + V_{k+1|k+1}P_{k+1|k+1}^b V_{k+1|k+1}^T \quad (27)$$

The design procedure of the TSKF is shown in Figure 2. The estimated system state from the “bias-free” estimator is fed into the compensation filter. The bias estimation from the “bias” filter is used to compensate for the bias-free system state estimation and to reconstruct the original system state.

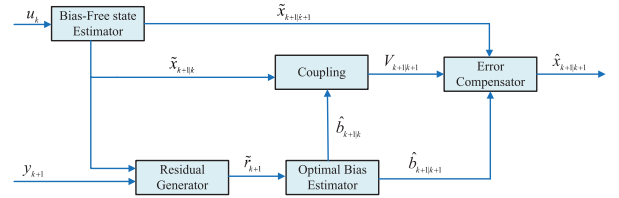


FIGURE 2. Design procedure of the TSKF.

IV. DESIGN OF ADRC CONTROLLER BASED ON TSKF

A. FAULT TOLERANT CONTROL STRUCTURE FOR QUADROTOR UAVS

When a quadrotor UAV actuator fails to function, there is a corresponding loss of actuator effectiveness. The four power units will be affected and the input to the quadrotor UAV power model will change accordingly. The loss-of-effectiveness fault factor from equation (7) is substituted into equation (3), and it has

$$\begin{bmatrix} u_z \\ u_\theta \\ u_\phi \\ u_\psi \end{bmatrix} = Q_1 \times \begin{bmatrix} (1-b_1)u_1 \\ (1-b_2)u_2 \\ (1-b_3)u_3 \\ (1-b_4)u_4 \end{bmatrix} \quad (28)$$

$$\text{where } Q_1 = \begin{bmatrix} K & K & K & K \\ KL & -KL & 0 & 0 \\ 0 & 0 & KL & -KL \\ KK_\psi & KK_\psi & -KK_\psi & -KK_\psi \end{bmatrix}$$

After simplification, u_1 , u_2 , u_3 and u_4 can be calculated using the following equation:

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = [Q_2 \ Q_3 \ Q_4 \ Q_5]^{-1} \begin{bmatrix} u_z \\ u_\theta \\ u_\phi \\ u_\psi \end{bmatrix} \quad (29)$$

where

$$Q_2 = \begin{bmatrix} K(1-b_1) \\ KL(1-b_1) \\ 0 \\ KK_\psi(1-b_1) \end{bmatrix}, Q_3 = \begin{bmatrix} K(1-b_2) \\ -KL(1-b_2) \\ 0 \\ KK_\psi(1-b_2) \end{bmatrix}, Q_4 = \begin{bmatrix} K(1-b_3) \\ 0 \\ KL(1-b_3) \\ -KK_\psi(1-b_3) \end{bmatrix}, Q_5 = \begin{bmatrix} K(1-b_4) \\ 0 \\ -KL(1-b_4) \\ -KK_\psi(1-b_4) \end{bmatrix}.$$

According to equation (5) and equation (29), the structural design of the quadrotor UAV fault tolerant control system is shown in Figure 3. According to the mathematical model of the quadrotor UAV, the system is divided into six channels with two loops. The position loop includes longitudinal motion channel (position x), lateral motion channel (position y), and height motion channel (height z). The attitude loop includes roll angle ϕ channel, pitch angle θ channel, and yaw angle ψ channel, respectively. Due to the cascaded structure of the longitudinal motion and pitch angle, and the lateral motion and roll angle, the model of the quadrotor UAV can be viewed as four relatively independent subsystems, namely the height subsystem, the yaw angle subsystem, the cascaded

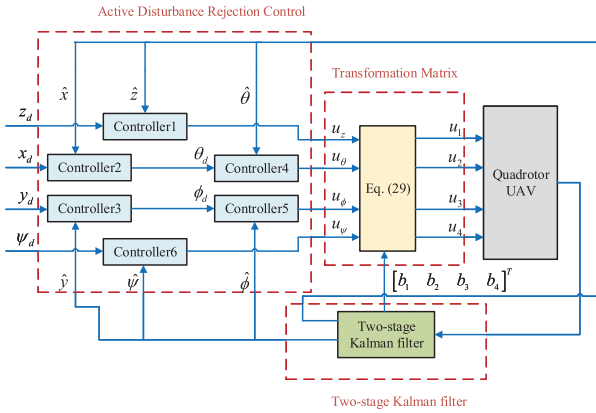


FIGURE 3. Fault tolerant control scheme for the quadrotor UAV.

subsystem of longitudinal motion and pitch angle, and the cascaded subsystem of lateral motion and roll angle.

B. DESIGN OF THE ADRC CONTROLLER

For the control structure shown in Figure 3, the coupling effects between loops can be estimated and compensated by ESO. With the compensation, the system can then be divided into six single-input and single-output (SISO) systems. Using the height subsystem as an example, the coupling effect and external disturbance are defined as the total disturbance f . By adding gravity compensation to the z axis controller, the altitude motion channel of the quadrotor UAV can be rewritten as

$$\ddot{y}(t) = f(t) + b_0 u(t) \quad (30)$$

where the total disturbance $f(t) = -a_1 \dot{y}(t) - a_2 y(t) + w(t) + (b_u - b_0)u(t)$, a_1 and a_2 are the model parameters, $w(t)$ is the sum of external disturbances and coupling effects, b_u and b_0 are the actual and estimated values of the control input gains. Assuming that the total disturbance f is differentiable and the first order derivative of f is h , equation (30) can be rewritten in the following form:

$$\begin{cases} \dot{x}(t) = Ax(t) + Bu(t) + Eh(t) \\ y(t) = Cx(t) \end{cases} \quad (31)$$

$$\text{where } A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, E = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, C = [1 \ 0 \ 0].$$

Based on equation (31), the ESO is designed as

$$\begin{cases} \dot{z}_1 = z_2 - \beta_1(z_1 - \hat{y}) \\ \dot{z}_2 = z_3 - \beta_2(z_1 - \hat{y}) + b_0 u \\ \dot{z}_3 = -\beta_3(z_1 - \hat{y}) \end{cases} \quad (32)$$

where z_1 , z_2 and z_3 are the estimated values of x_1 , x_2 and x_3 , respectively. β_1 , β_2 and β_3 are the error feedback gains of the ESO. The original y is replaced by $\hat{y}$ estimated from the TSKF. The characteristic equation of the ESO is expressed

$$\lambda(s) = s^3 + \beta_1 s^2 + \beta_2 s + \beta_3 \quad (33)$$

Place the closed-loop system poles at $-\omega_o$, equation (33) can be rewritten as $\lambda_o(s) = (s + \omega_o)^3$. The three error feedback gains of the ESO are obtained as follows:

$$\begin{cases} \beta_1 = 3\omega_o \\ \beta_2 = 3\omega_o^2 \\ \beta_3 = \omega_o^3 \end{cases} \quad (34)$$

where ω_o is the ESO bandwidth. With the pole configuration, the ESO parameters are reduced to one, which greatly simplifies the design of ESO. By adjusting ω_o , z_1 , z_2 and z_3 can estimate the values of y , $\dot{y}$ and f accurately.

The linear state error feedback control law for ADRC is designed as

$$\begin{cases} u_0 = k_p(r - z_1) + k_d(\dot{r} - z_2) + \ddot{r} \\ u = (u_0 - z_3)/b_0 \end{cases} \quad (35)$$

where r is the reference signal, k_p is the proportional factor and k_d is the differential factor.

Place the closed-loop system poles at $-\omega_c$, equation (33) can be rewritten as $\lambda_c(s) = (s + \omega_c)^2$, and controller parameters can be obtained as $k_p = \omega_c^2$ and $k_d = 2\omega_c$.

C. STABILITY ANALYSIS

TSKF can accurately estimate the loss-of-effectiveness fault factor of the system when the total disturbance of the system is mainly composed of external disturbances. Assuming that the derivatives of the total disturbances are bounded, the convergence of ESO and the stability of the closed-loop system are guaranteed by the following theorems.

Theorem 1: Assuming h is bounded, $|h| \leq \delta$, there exist positive constants $\sigma_i > 0$ ($i = 1, 2, 3$) and a finite time $T > 0$ such that $|x_i(t) - z_i(t)| \leq \sigma_i$.

Proof: [Proof of Theorem 1] According to equation (32) and equation (34), the error form of the ESO is written as

$$\dot{\varepsilon}(t) = \omega_o A_e \varepsilon(t) + B_e h / \omega_o^2 \quad (36)$$

$$\text{where } \varepsilon(t) = [\varepsilon_1(t) \ \varepsilon_2(t) \ \varepsilon_3(t)]^T, \varepsilon_i(t) = (x_i(t) - z_i(t)) / \omega_o^{i-1}, A_e = \begin{bmatrix} -3 & 1 & 0 \\ -3 & 0 & 1 \\ -1 & 0 & 0 \end{bmatrix}, B_e = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Solving equation (36), one obtains

$$\varepsilon(t) = e^{\omega_o A_e t} \varepsilon(0) + \int_0^t e^{\omega_o A_e (t-\tau)} B_e h / \omega_o^2 d\tau \quad (37)$$

Let $q(t) = \int_0^t e^{\omega_o A_e (t-\tau)} B_e h / \omega_o^2 d\tau$, since $|h| \leq \delta$, then it has

$$|q_i(t)| \leq \frac{\delta}{\omega_o^3} \left(\left| (A_e^{-1} B_e)_i \right| + \left| (A_e^{-1} e^{\omega_o A_e t} B_e)_i \right| \right) \quad (38)$$

Since A_e is Hurwitz, there exists a finite time $T > 0$ that

$$\begin{cases} (e^{\omega_o A_e t})_{ij} \leq \frac{1}{\omega_o^3} \\ (e^{\omega_o A_e t} B_e)_i \leq \frac{1}{\omega_o^3} \end{cases} \quad (39)$$

According to the expressions of A_e and B_e , equation (38) can be further expressed as

$$|q_i(t)| \leq \frac{\delta}{\omega_o^3} + \frac{4\delta}{\omega_o^6} \quad (40)$$

With equation (37) and equation (39), one obtains

$$\left| \left(e^{\omega_o A_e t} \varepsilon(0) \right)_i \right| \leq \frac{\sum_{i=1}^3 |\varepsilon_i(0)|}{\omega_o^3} \quad (41)$$

Combining equation (37), equation (40) and equation (41), one has

$$|\varepsilon_i(t)| \leq \frac{\sum_{i=1}^3 |\varepsilon_i(0)|}{\omega_o^3} + \frac{\delta}{\omega_o^3} + \frac{4\delta}{\omega_o^6} \quad (42)$$

Owing to $\varepsilon_i(t) = (x_i(t) - z_i(t)) / \omega_o^{i-1}$, there is

$$|x_i(t) - z_i(t)| \leq \frac{\sum_{i=1}^3 \omega_o^{1-i} |x_i(0) - z_i(0)|}{\omega_o^{4-i}} + \frac{\delta}{\omega_o^{4-i}} + \frac{4\delta}{\omega_o^{7-i}} = \sigma \quad (43)$$

From equation (43), it can be seen that the estimation error of the ESO is bounded and its upper bound decreases with the increase of the observer bandwidth. ■

Theorem 2: Assuming h is bounded, there exist positive constants $\omega_o > 0$ and $\omega_c > 0$, that the closed-loop system is stable.

Proof: [Proof of Theorem 2] Let $e_{c1} = r - x_1$ and $e_{c2} = \dot{r} - \dot{x}_2$, then one has

$$\dot{e}_{c1} = \dot{r} - \dot{x}_1 = e_{c2} \quad (44)$$

$$\dot{e}_{c2} = \ddot{r} - \ddot{x}_2 = \ddot{r} - (x_3 + b_0 u) \quad (45)$$

According to equation (35), equation (45) can be expressed as

$$\begin{aligned} \dot{e}_{c2} &= \ddot{r} - x_3 - [k_p(r - z_1) + k_d(\dot{r} - z_2) - z_3 + \ddot{r}] \\ &= -k_p e_{c1} - k_d e_{c2} - k_p \varepsilon_1 - k_d \omega_o \varepsilon_2 - \omega_o^2 \varepsilon_3 \end{aligned} \quad (46)$$

Combining equation (44) and equation (46), one has

$$\dot{e}_c(t) = A_c e_c(t) + B_c \varepsilon(t) \quad (47)$$

where

$$A_c = \begin{bmatrix} 0 & 1 \\ -k_p & -k_d \end{bmatrix}, \quad B_c = \begin{bmatrix} 0 & 0 & 0 \\ -k_p & -k_d \omega_o & -\omega_o^2 \end{bmatrix}.$$

According to Theorem 1, ε_i ($i = 1, 2, 3$) is going to zero with the increase of ω_o , therefore it has $\lim_{t \rightarrow \infty} \|B_c \varepsilon(t)\| \rightarrow 0$. Since A_c is Hurwitz, then there exists $\omega_o > 0$ and $\omega_c > 0$ that $\lim_{t \rightarrow \infty} e_{ci}(t) \rightarrow 0$ ($i = 1, 2$) and the closed-loop system is stable. ■

V. SIMULATION AND DISCUSSION

In this section, comparative experiments are designed to verify the performance and robustness of the proposed fault tolerant control scheme (KF-ADRC). To highlight its performance, a conventional ADRC scheme and a PID scheme are used for comparison. In the trajectory tracking

experiment, the initial position of the quadrotor UAV is $[0, 0, 0]$ m, the initial value of the angle is $[0, 0, 0]$ rad, and the sampling time is 0.001s. The parameters of the ADRC, PID, and KF-ADRC are shown below.

PID parameters:

Position loop: $P_x = 0.104$, $I_x = 0.012$, $D_x = 0.205$, $P_y = -0.1$, $I_y = -0.1$, $D_y = -0.4$, $P_z = 0.3$, $I_z = 0.01$, $D_z = 1.1$.

Attitude loop: $P_\theta = 2.353$, $I_\theta = 0.558$, $D_\theta = 0.902$, $P_\phi = 2.353$, $I_\phi = 0.558$, $D_\phi = 0.902$, $P_\psi = 2.353$, $I_\psi = 0.558$, $D_\psi = 0.902$.

ADRC parameters:

Position loop: $\omega_o = 30$, $\omega_c = 2$, $b_0 = 1$.

Attitude loop: $\omega_o = 60$, $\omega_c = 30$, $b_0 = 1$.

KF-ADRC parameters:

Position loop: $\omega_o = 30$, $\omega_c = 2$, $b_0 = 1$.

Attitude loop: $\omega_o = 60$, $\omega_c = 30$, $b_0 = 1$.

In order to evaluate its performance, the absolute error integral (IAE) index is used in the simulation experiments:

$$IAE = \int_0^T |r(t) - y(t)| dt \quad (48)$$

A. SETPOINT TRACKING EXPERIMENT WITH MEASUREMENT NOISE

The purpose of this experiment is to verify the closed-loop dynamic performance of the KF-ADRC control scheme in the presence of measurement noise. White noise is added to the position and attitude loops.

$$Q_k^x = \begin{bmatrix} 10^{-6} \times I_{6 \times 6} & 0_{6 \times 6} \\ 0_{6 \times 6} & 10^{-12} \times I_{6 \times 6} \end{bmatrix}, \quad R_{k+1} = \begin{bmatrix} 10^{-6} \times I_{3 \times 3} & 0_{3 \times 3} \\ 0_{3 \times 3} & 10^{-12} \times I_{3 \times 3} \end{bmatrix}$$

and $Q_k^b = 10^{-1} \times I_{4 \times 4}$. The reference signal of the position loop is $(1, 1, 1)$ m, and the set point of the yaw angle is $\psi_d = 1$. Since the pitch angle and roll angle are indirect control quantities, it is not possible to set the reference value directly. Therefore, only the IEA index of the four channels (x, y, z, ψ) is compared. The simulation results of the position loop and angle loop are shown in Figure 4 and Figure 5, and the IEA index is shown in Table 2.

It can be seen that the ADRC scheme and the KF-ADRC scheme achieve better tracking performance in the position and attitude loops. But the PID scheme has a large amount of overshoot. In the x, y, z and ψ loops, the IEA index of the ADRC scheme is close to that of the KF-ADRC scheme, so only the IEA index of the KF-ADRC scheme and the PID scheme is compared below. The IEA index of the KF-ADRC scheme is reduced by 5.67% compared to the PID scheme for the x loop. In the y loop, the IEA index of the KF-ADRC scheme is 23.85% lower than that of the PID scheme. In the z loop, the IEA index of the KF-ADRC scheme is reduced by 50.74% compared to the PID scheme. In the ψ loop, the

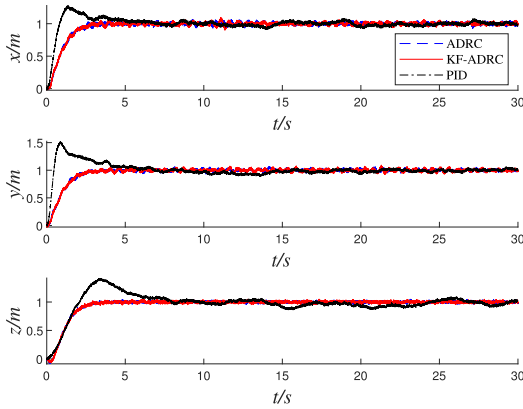


FIGURE 4. Position loop setpoint tracking with measurement noise.

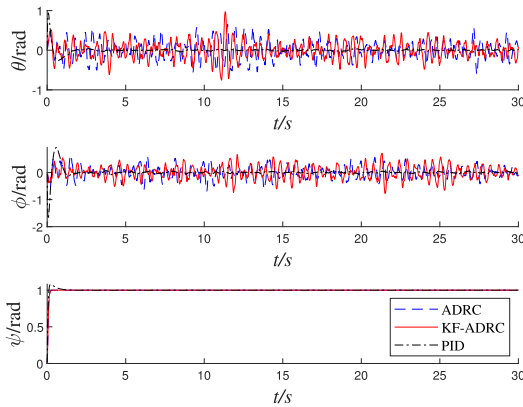


FIGURE 5. Attitude loop setpoint tracking with measurement noise.

TABLE 2. IAE index for different schemes with measurement noise.

	ADRC	KF-ADRC	PID
IAE for x loop	1.4182	1.4177	1.5030
IAE for y loop	1.3891	1.4589	1.9159
IAE for z loop	1.5518	1.5527	3.1526
IAE for ψ loop	0.0677	0.0677	0.6757

IEA index of the KF-ADRC scheme is reduced by 89.98% compared to the PID scheme.

The results show that the ADRC scheme and the KF-ADRC scheme have the advantages of short settling time and no overshoot when only measurement noise is added in the loop. Therefore, the control performance of the ADRC scheme and the KF-ADRC scheme is better than that of the PID scheme in the presence of measurement noise.

B. SETPOINT TRACKING EXPERIMENT WITH MEASUREMENT NOISE AND LOSS-OF-EFFECTIVENESS FAULT

The loss-of-effectiveness fault factors is set as $[0.4 \ 0.5 \ 0.4 \ 0.5]^T$. The position loop reference signal is $(1, 1, 1)$ m, and the set point of the yaw angle is $\psi_d = 1$. The actuator loss-of-effectiveness fault factors estimated by the TSKF are shown in Figure 6 - Figure 9. The simulation results of the position loop and attitude loop are shown in Figure 10 and Figure 11, and the IAE index is shown in Table 3.

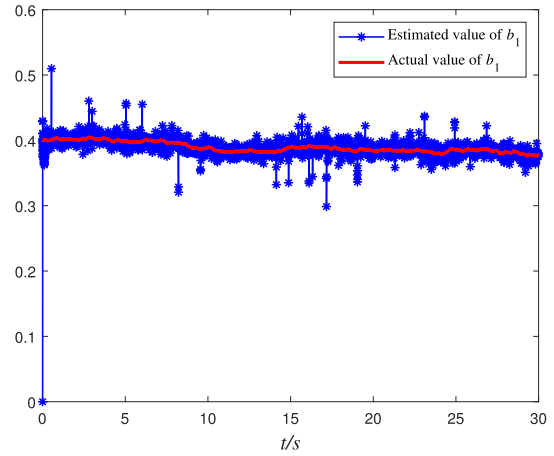


FIGURE 6. Comparison between the estimated value and the actual value of b_1 .

TABLE 3. IAE index for different schemes with measurement noise and loss-of-effectiveness fault.

	ADRC	KF-ADRC	PID
IAE for x loop	1.5017	1.4228	5.7214
IAE for y loop	1.4707	1.4580	5.2929
IAE for z loop	1.8892	1.5583	734.3833
IAE for ψ loop	0.2043	0.0677	0.1339

From Figure 6 - Figure 9, it can be seen that the TSKF is able to accurately estimate the loss-of-effectiveness fault factors. Only a very few data points deviate from the actual value. These values can be compensated by ESO, which reduces the impact on the fault-tolerant control of the quadrotor UAV. Therefore, TSKF can be effectively applied to fault tolerant control.

The tracking simulation of the position loop shows that the ADRC scheme and the KF-ADRC scheme have superior tracking performance in the x and y loops, while the PID scheme has a large amount of overshoot. In the z loop, the KF-ADRC scheme can track the step signal quickly and accurately. The ADRC scheme converges relatively slower, while the PID scheme drops directly due to gravity in the loop.

For the simulation of the attitude loop, only the KF-ADRC scheme is compared here with the ADRC scheme because the PID scheme in the position loop drops directly in the z loop, and loses stability. In the θ and ϕ loops, the KF-ADRC scheme is more stable and has a smaller amplitude than the ADRC scheme, while the amplitude value of the ADRC scheme is close to ± 1 and has larger amplitude. In the ψ loop, both the KF-ADRC scheme and the ADRC scheme are fast tracking, but the ADRC scheme has a larger overshoot. Table 3 shows that the IEA index of KF-ADRC is better than that of the ADRC scheme in the x , y , z and ψ loops. In the ψ loop, the IEA index of the KF-ADRC scheme decreases most significantly, and the IEA index of the KF-ADRC scheme is 66.86% lower than that of the ADRC scheme. The results show that the KF-ADRC scheme has better stability in the presence of measurement noise and actuator loss-of-effectiveness fault.

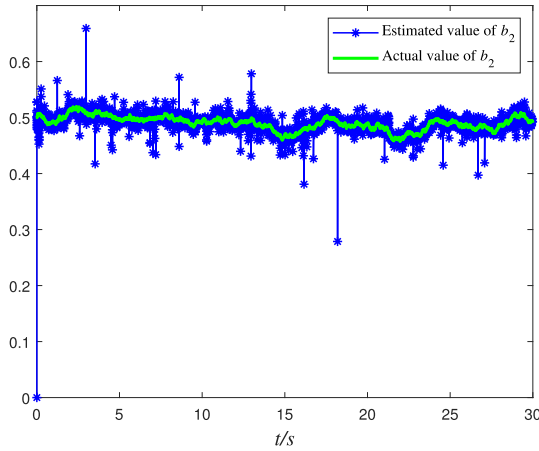


FIGURE 7. Comparison between the estimated value and the actual value of b_2 .

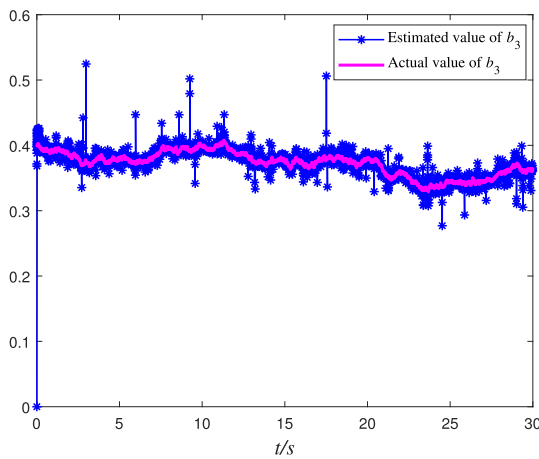


FIGURE 8. Comparison between the estimated value and the actual value of b_3 .

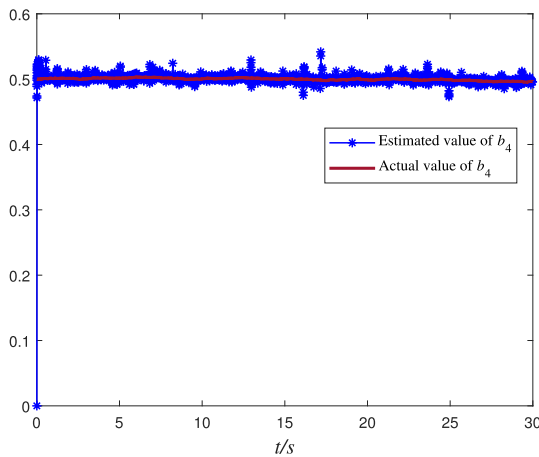


FIGURE 9. Comparison between the estimated value and the actual value of b_4 .

C. DISTURBANCE REJECTION EXPERIMENT

To further verify the performance of the proposed scheme, external disturbances are added to the simulation experiment. The spiral trajectory tracking simulation is performed. The model, noise conditions, and faults are the same as in the

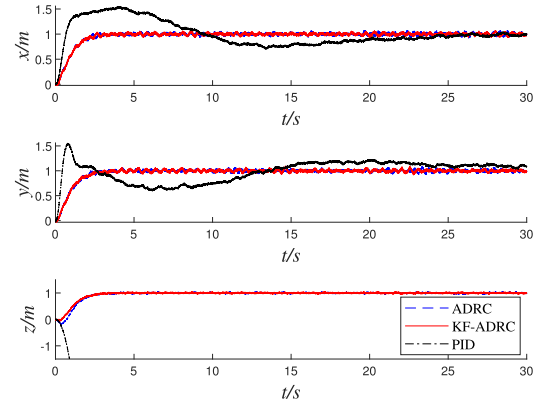


FIGURE 10. Position loop with measurement noise and loss-of-effectiveness fault.

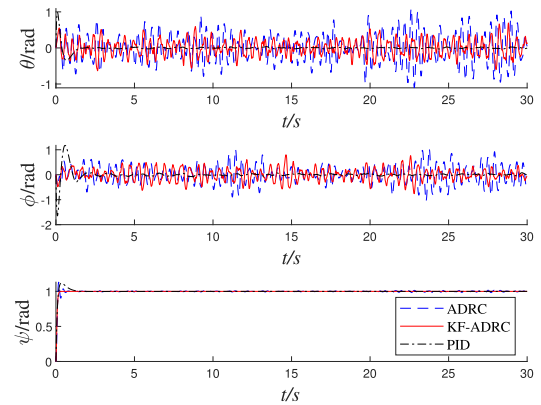


FIGURE 11. Attitude loop with measurement noise and loss-of-effectiveness fault.

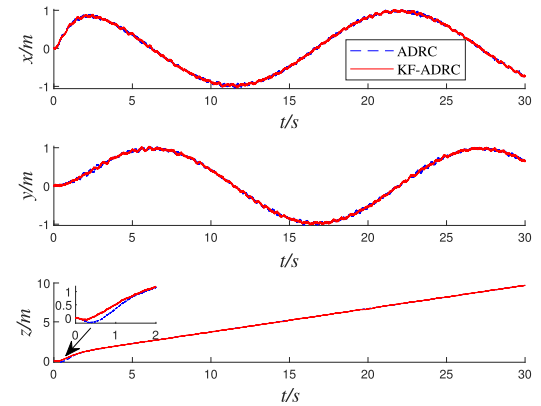


FIGURE 12. Position loop with external disturbance.

previous experiment. The position loop reference signal is $(1, 1, 1)$ m, and the set point of the yaw angle is $\psi_d = 1$. The simulation results are shown in Figure 12, Figure 13 and Figure 14, and the IEA index is shown in Table 4. Since the PID scheme is dropped due to gravity after adding the fault in the previous experiment, the PID scheme is not considered here. In the x and y loops of Figure 12, the amplitude of the ADRC scheme is larger than that of the KF-ADRC scheme, and there is a position tracking error in the ADRC scheme at the initial position of the z loop. In Figure 13, the amplitude of the ADRC scheme has exceeded $\pm 60^\circ$ in the attitude loops.

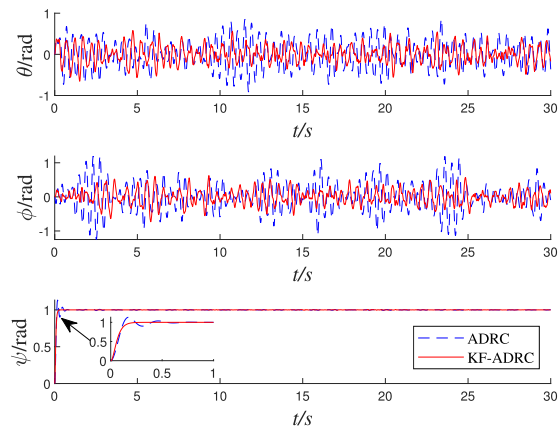


FIGURE 13. Attitude loop with external disturbance.

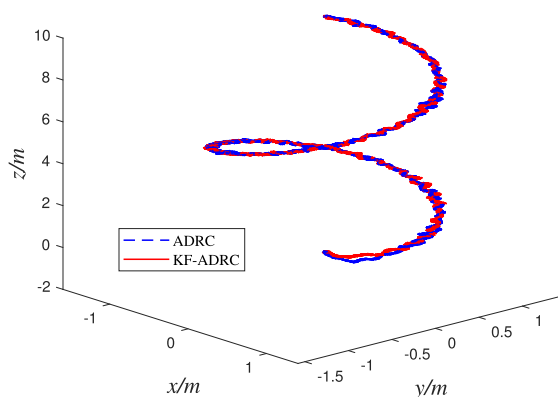


FIGURE 14. Comparison of 3D trajectories of different control schemes.

TABLE 4. IAE index for different schemes with external disturbance.

	ADRC	KF-ADRC
IAE for x loop	6.4439	6.4326
IAE for y loop	5.3511	5.3364
IAE for z loop	10.2807	10.0655
IAE for ψ loop	0.1488	0.0882

In contrast, the KF-ADRC scheme has a smaller amplitude and is relatively stable. In the ψ loop, the ADRC scheme has large overshoot.

The IEA index of the KF-ADRC scheme is close to those of the ADRC scheme in the x , y and z loops. However, in the ψ loop, the KF-ADRC scheme showed the most significant decrease in the IEA index, with a 40.72% reduction compared to the ADRC scheme. Figure 14 shows the 3D trajectory tracking of the quadrotor position loop drawn from Figure 12. From Figure 14, it can be seen that the KF-ADRC scheme can track the reference signal more quickly and accurately when the external disturbance is added. As shown above, the KF-ADRC scheme has better stability and disturbance rejection performance.

VI. CONCLUSION

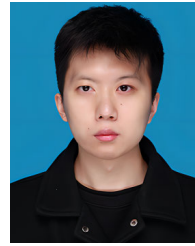
In this paper, a cascade ADRC strategy with a TSKF is proposed for a quadrotor UAV with the actuator

loss-of-effectiveness fault and measurement noise. Combining ADRC and TSKF allows the controller to respond promptly to faults and effectively compensate for the impact of external disturbances on fault coefficient estimation through ESO. The stability analysis shows that the estimation and closed-loop tracking errors are guaranteed to remain bounded. The simulation results demonstrate the effectiveness of the proposed method. In comparison to the PID control approach, the proposed method guarantees stable flight even in the case of a fault. The IAE index is reduced by a minimum of 49.43%. Furthermore, the proposed strategy exhibits superior stability in the attitude loop compared to the traditional ADRC strategy. Specifically, the pitch angle fluctuation of the proposed strategy is 40.72% lower than that observed with the ADRC strategy, and the roll angle fluctuation is 28.83% lower than that observed with the ADRC strategy. As a result, the proposed strategy shows great potential for reliable flight control of quadrotor UAVs.

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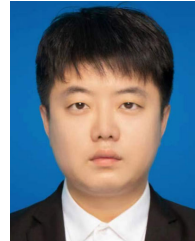
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YUSHENG DU was born in Jiangsu, China, in 1999. He received the B.E. degree in automation from Nantong University, Nantong, China, in 2021, where he is currently pursuing the bachelor's degree.

His research interests include active disturbance rejection control and Kalman filter theory.



PENG HUANG was born in Anhui, China, in 1998. He received the B.E. degree in automation from Nantong University, Nantong, China, in 2020, where he is currently pursuing the bachelor's degree. His research interests include UAV flight control and active disturbance rejection control.



YUN CHENG was born in Jiangsu, China, in 1990. He received the B.E. degree in automation from the Nanjing University of Science and Technology, Nanjing, China, in 2013, the M.E. degree in control science and engineering from the Tianjin University of Technology, Tianjin, China, in 2016, and the Ph.D. degree in control science and engineering from Nankai University, Tianjin, in 2019.

He is currently a Lecturer with the School of Electrical Engineering, Nantong University. His

research interests include active disturbance rejection control, intelligent predictive control, and robust control.



YUNLEI FAN was born in Jiangsu, China, in 1998. He received the B.E. degree in automation from the Xinglin College, Nantong University, Nantong, China, in 2021, where he is currently pursuing the bachelor's degree.

His research interests include active disturbance rejection control and process control.



YINLONG YUAN was born in Henan, China, in 1990. He received the B.E. degree in electrical engineering and automation from Henan Polytechnic University, Jiaozuo, China, in 2013, and the Ph.D. degree in pattern recognition and intelligent system from the South China University of Technology, Guangzhou, China, in 2019.

He has been a Lecturer with the College Electrical Engineering, Nantong University, Nantong, China, since 2019. His research interests include deep reinforcement learning, UAVs, and Bayesian deep learning.

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